

## DSA0602 Data Handling and Visualization

**NAME : Shri Harshini Devi . A**

**REG.NO : 192524104**

**Dept. Name : B.Tech-AIDS**

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**Q1. A hospital wants to tell patients their risk of a rare surgical complication. Version A states "there is a 2% chance of complication." Version B states "out of 100 patients like you, 2 will experience a complication." Evaluate which framing is more likely to be correctly understood by a layperson, and justify your judgment using at least one known pitfall of probability communication. Under what circumstances would you recommend switching to the other framing?**

### **Q1. Probability Communication and Framing**

**Version B is more likely to be correctly understood by a layperson.**

- **Version A:** "There is a 2% chance of complication."
- **Version B:** "Out of 100 patients like you, 2 will experience a complication."

Version B uses a **natural-frequency format**, which is generally easier for people to understand than abstract percentages. A known pitfall in probability communication is **denominator neglect/base-rate confusion**: people may understand "2%" but fail to visualize what it means in terms of actual patients.

Version B gives a concrete reference group of 100 patients, making the risk easier to interpret. It also reduces the possibility that patients misunderstand 2% as "2 out of every 10" or perceive it as either extremely large or negligible without context.

**Recommendation:** Use Version B for routine patient communication, especially when explaining risk to patients with limited statistical knowledge.

**Switch to Version A** when:

- communicating with medically trained or statistically literate patients,
- comparing several risks expressed consistently as percentages,
- or when the risk is extremely small and a percentage provides greater precision.

**Conclusion:** For general patients, **natural frequencies such as “2 out of 100” are clearer and more intuitive than percentages.**

**Q2. A weather agency is deciding between two ways to show forecast uncertainty for tomorrow's rainfall: (a) a static confidence-interval band, or (b) an animated Hypothetical Outcome Plot cycling through simulated draws. Evaluate both options for a general public audience viewing on a mobile app, weighing comprehension speed, risk of misinterpretation, and technical feasibility. Defend your final recommendation.**

**Q2. Forecast Uncertainty: Static Band vs. Animated HOP**

The two approaches have different strengths.

| Criterion              | Static Confidence-Interval Band | Animated Hypothetical Outcome Plot           |
|------------------------|---------------------------------|----------------------------------------------|
| Comprehension speed    | <b>Fast</b>                     | Slower                                       |
| Public familiarity     | <b>Higher</b>                   | Lower                                        |
| Shows uncertainty      | Good                            | <b>Very intuitive when understood</b>        |
| Misinterpretation risk | Moderate                        | Higher if animation is ignored/misunderstood |
| Mobile suitability     | <b>Excellent</b>                | Moderate                                     |
| Technical complexity   | <b>Low</b>                      | Higher                                       |

| Criterion     | Static Confidence-Interval Band | Animated Hypothetical Outcome Plot             |
|---------------|---------------------------------|------------------------------------------------|
| Accessibility | Easier                          | Can create problems for motion-sensitive users |

A **static confidence-interval band** immediately shows the expected rainfall and the range of uncertainty. It requires little processing and works well on a small mobile screen.

An **animated Hypothetical Outcome Plot (HOP)** cycles through simulated possible rainfall outcomes. This can help users understand that uncertainty represents many possible outcomes rather than a single fixed prediction. However, animation may be too fast, may distract users, and may be difficult for people who do not understand probabilistic graphics. It also requires more technical implementation and accessibility considerations.

**Final recommendation:** Choose the **static uncertainty band** for the main mobile weather application.

It provides the best balance of **speed, simplicity, accessibility, and technical feasibility**. An HOP could be offered as an optional interactive explanation for users who want more detail.

**Conclusion:** For a general public mobile audience, **static visualization should be the primary choice; animation should be supplementary rather than mandatory.**

**Q3. A startup's pitch deck shows revenue growth from ₹2 lakhs to ₹2 crores over 3 years using a bar chart on a linear axis, with the last bar's height exaggerated by a non-zero baseline. Critique this chart against the principle of proportional ink and the choice of axis scale. Evaluate whether a corrected version would strengthen or weaken the startup's growth story, and justify your conclusion.**

### **Q3. Revenue Growth, Non-Zero Baseline and Proportional Ink**

The original chart is misleading for two major reasons.

First, the revenue increases from **₹2 lakhs to ₹2 crores**, which represents a very large increase. A **linear axis** is appropriate because revenue is a quantitative variable and the visual height should be proportional to the actual values.

However, using a **non-zero baseline** exaggerates the apparent height difference between the bars. The visual difference becomes larger than the actual numerical difference represented by the data. This violates the principle of **proportional ink**: the amount of visual emphasis should accurately correspond to the underlying quantities.

For example:

- ₹2 lakhs = ₹0.02 crore
- ₹2 crores = ₹2 crore

Therefore, revenue increased by approximately **100 times**.

A corrected chart should:

1. Use a **linear y-axis**.
2. Start the bar axis at **zero**.
3. Use equal intervals on the axis.
4. Keep bar heights proportional to revenue.
5. Clearly label the values.

The corrected chart would **strengthen rather than weaken the startup's growth story**.

Although the exaggerated visual effect disappears, the actual data already demonstrate extremely strong growth. Presenting the growth honestly makes the pitch more credible and trustworthy.

**Conclusion:** A zero-baseline linear bar chart provides a more accurate and defensible representation. The startup's growth remains impressive because the **data itself—not graphical distortion—supports the claim**.

**Q4.** A city government's COVID dashboard uses a rainbow ("jet") color scale to show infection rates by district, with red-green used elsewhere to flag "safe" vs. "high-risk" zones. Evaluate this design against three criteria: perceptual accuracy, accessibility for color-vision-deficient users, and public trust during a health emergency. Rank the two color choices by severity of harm and justify your ranking.

#### Q4. Rainbow Scale vs. Red-Green Health Dashboard

The two color choices should be evaluated differently.

##### 1. Rainbow/Jet color scale

A rainbow scale is problematic because its colors do not correspond consistently to perceived numerical differences. Some transitions appear much more visually prominent than others even when the numerical differences are equal.

This reduces **perceptual accuracy** and can cause users to perceive artificial boundaries in infection rates.

It is also problematic for **color-vision-deficient users**, because several colors in the rainbow spectrum can be difficult to distinguish.

##### 2. Red-Green safe/high-risk scale

Red-green encoding is intuitive because people commonly associate:

- Green → safe
- Red → danger/high risk

However, approximately 8% of men and a smaller percentage of women have some form of color-vision deficiency, making red-green distinctions difficult for many users.

This can become particularly dangerous in a **health emergency**, because a user may fail to distinguish a safe district from a high-risk district.

#### Ranking by severity of harm

##### 1. Red-green scale — More severe harm

This should rank first because it directly assigns critical health information—**safe versus high-risk**—to a color distinction that some users cannot reliably perceive. A mistaken interpretation could affect real-world decisions.

##### 2. Rainbow/jet scale — Serious but comparatively less harmful

Jet creates perceptual distortion and accessibility problems, but its primary problem is that it can exaggerate or obscure differences rather than directly making two critical categories indistinguishable.

#### Recommendation

Replace both with an **accessible sequential color scale**, such as light-to-dark single-hue shading, for infection rates. For categorical risk levels, combine color with **labels, symbols, or patterns** so that color is not the only information channel.

**Conclusion:** The red-green design has the greater immediate safety risk, while the rainbow scale has major perceptual and accessibility problems. Both should be replaced.

**Q5. An analyst must visualize 50,000 GPS pings to show where a ride-hailing app's demand concentrates in a city. Three proposals are on the table: (a) a plain scatter plot, (b) a 2D density contour map, (c) a direct-area bubble map aggregated by neighborhood. Evaluate the trade-offs of each approach for both accuracy and interpretability, and justify which one you would approve for the company's operations dashboard.**

**Q5. Visualizing 50,000 GPS Pings**

The three approaches have different trade-offs.

| Method                     | Accuracy                       | Interpretability       | Main Advantage                | Main Problem                                    |
|----------------------------|--------------------------------|------------------------|-------------------------------|-------------------------------------------------|
| A. Plain scatter plot      | High for individual points     | Low with 50,000 points | Shows actual locations        | Severe overplotting                             |
| B. 2D density contour map  | High for spatial concentration | High                   | Clearly shows demand hotspots | Exact individual locations are less visible     |
| C. Neighborhood bubble map | Moderate                       | High                   | Easy operational comparison   | Aggregation hides within-neighborhood variation |

**A. Plain Scatter Plot**

A scatter plot preserves the location of every GPS ping, so it has high **data-level accuracy**. However, with 50,000 points, severe **overplotting** will occur.

Thousands of points may overlap, making dense areas appear as dark blobs. Individual observations become difficult to distinguish, and the chart can become visually cluttered.

**Verdict:** Accurate but poorly interpretable for an operations dashboard.

## B. 2D Density Contour Map

A density map aggregates nearby points into spatial density. It is particularly suitable for identifying **hotspots and demand concentrations**.

It reduces overplotting and makes spatial patterns much easier to interpret. Operations managers can quickly identify areas where ride demand is concentrated.

The disadvantage is that individual GPS observations are no longer directly visible.

**Verdict:** Best balance of spatial accuracy and interpretability for understanding demand concentration.

## C. Neighborhood Bubble Map

This method aggregates demand by neighborhood and represents the total using bubble size.

It is highly understandable and useful for comparing neighborhoods. However, it introduces **aggregation bias**: two neighborhoods with the same total demand may have very different internal spatial patterns.

Large neighborhoods can also visually dominate smaller ones if raw counts are used without considering area or population.

**Verdict:** Excellent for high-level neighborhood comparison, but less accurate for detailed spatial demand analysis.

## Final Recommendation

I would approve **Option B: the 2D density contour map** for the company's operations dashboard.

The main business question is “**where does demand concentrate?**”, not “where exactly is every individual GPS ping?” A density map directly answers this question while avoiding the severe overplotting problem of the scatter plot.

For an operations dashboard, I would also add:

- zoom and pan,
- a time filter such as hour/day,
- a clear density legend,
- and optionally neighborhood boundaries.

## Final conclusion:

**2D density contour map > neighborhood bubble map > plain scatter plot** for this specific

operational purpose because it provides the strongest combination of **accuracy, scalability, and rapid interpretation**.